

Hierarchical Context Guides Human Memory Search

Charlotte A. Cornell¹ and Qiong Zhang^{1,2,3}

¹ Psychology Department, Rutgers University–New Brunswick

² Computer Science Department, Rutgers University–New Brunswick

³ Rutgers Center for Cognitive Science, Rutgers University–New Brunswick

Abstract

A fundamental property of human memory is its ability to organize a continuous stream of information into discrete, searchable events. Although empirical work supports hierarchical memory representations in the brain, we lack an explicit theory of how humans search this structure. Here, we present a hierarchical model of memory search that unifies a wide range of phenomena under a single framework. By tracking mental contexts at multiple levels, the model captures memory behavior across different timescales, from recalling single lists of information to navigating materials consisting of a series of lists. It explains enhanced recall at list boundaries by proposing a top-down hierarchical search: higher-level contextual cues are retrieved before activating lower-level item information. This mechanism also accounts for recall transition patterns across list boundaries, reproducing effects observed in the event memory literature. Taken together, these results provide a unified account of how humans efficiently organize and search their memories.

Keywords: memory search, free recall, hierarchical memory, event memory, computational modeling

Hierarchical Context Guides Human Memory Search

Human experience unfolds as a continuous stream, yet we process this flow as distinct and meaningful events^{1,2}. Despite the potential for overlap and interference among many memories, people can focus on a specific part of their past and recall it in detail³. Not only can we recall recent information but also mentally travel back in time to access remote memories^{4,5}. How do we efficiently search such a vast store of memories? In this work, we propose that humans do so by organizing and searching their memories hierarchically.

Past research supports a hierarchical representation of memories in the brain. This is seen in event segmentation tasks where people parse continuous stimuli into discrete, nested events^{1,6,7}. For instance, when people segmented and described video clips in coarse or fine units, the coarse boundaries tended to overlap with fine ones, revealing an underlying hierarchical structure⁸. Neuroscience work suggests the brain mirrors this structure by processing information across multiple timescales. Studies using functional magnetic resonance imaging (fMRI) and electrocorticographic (ECoG) recording have found that low-level sensory areas track rapid moment-to-moment changes spanning a few milliseconds, while high-level areas track slower changes spanning seconds to minutes (using movie^{9,10} and audiobook¹¹ stimuli scrambled at different timescales). Moreover, fMRI studies with movie-viewing tasks have observed that relatively stable patterns of activity at different cortical levels form a partially nested hierarchy^{12,13} and that shifts in activations of these regions align with human-rated event boundaries¹². This multiscale representation is further supported by animal studies using single-neuron recordings, which have showed that different hippocampal neurons fire at different moments in temporally structured experiences^{14,15}.

Despite this rich behavioral and neural data suggesting hierarchical organization, there lacks computationally explicit theories that specify how humans search their memory hierarchically. Previous computational modeling of memory search has predominantly focused on materials consisting of a single list of items, using list learning paradigms such as the free recall task. In a free recall task, participants study a list of items and recall as many items from the list as possible in any order^{16,17} (Figure 1c). A prominent memory search model, the Context Maintenance and Retrieval (CMR) model^{18–20}, explains these item-level effects by organizing a single list of items into a latent context space based on study order and using the current context state to cue retrieval. While the model accounts for recency (enhanced recall for end-of-list items¹⁶) and temporal contiguity (successive recalls tend to come from nearby serial positions²¹) at the level of items in one list, there is evidence showing that these effects also emerge at a higher, hierarchical level. When the materials involve a series of lists, recency and temporal contiguity effects are shown to be scale-invariant, extending from short intervals within a list to longer intervals across lists (i.e., a tendency to recall more items from later lists and to transition to lists studied nearby^{22,23}). Furthermore, the context-based account alone in CMR is not sufficient to explain additional recall patterns at list boundaries, such as primacy effects (enhanced recall of start-of-list items¹⁶). This primacy effect occurs for recall of recent information (a just-studied list¹⁶) and remote information (lists studied before the most just-studied one²⁴). Together, this literature highlights the need to integrate existing models of memory search with mechanisms of hierarchical search to capture the full range of memory behavior.

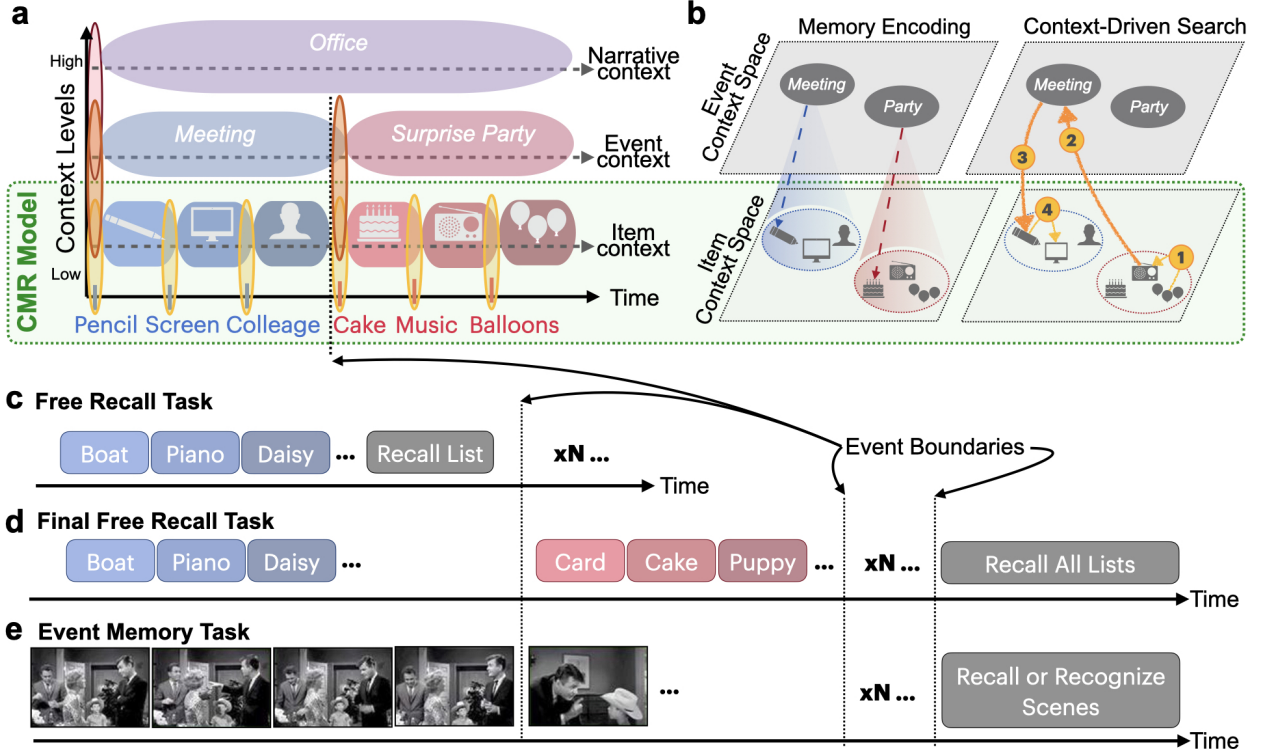


Figure 1

Illustrations of the Context Maintenance and Retrieval (CMR) model^{18–20}, our proposed hierarchical model, and related memory tasks. (a) CMR tracks a single, item-level context that drifts towards each new item and to which items are associated (yellow ovals). Our hierarchical model maintains multiple levels of context that evolve at different rates. In addition to associations forming between items and item-level context (yellow ovals), our model resets and synchronizes the item-level context to the event-level context at event boundaries, with the event-level context encoded in memory (orange ovals). (b) The encoding processes organize items into the item context space based on study order, and in our model, the links the event-level context to the item-level context at boundaries (dashed lines). Therefore, at recall, (1) the current item-level context drives search, and in our model, (2) the event-level context can be used to (3) reactivate the item-level context at a boundary, (4) which drives further search. (c) Formulated this way, our model captures hierarchical behavior in list learning tasks of free recall, in which a single list of items is studied and then recalled in any order¹⁶, and (d) final free recall, in which multiple lists are studied and then all items are recalled in any order²⁴. (e) Our model also captures enhanced recall of start-of-list items and a stronger tendency to transition within lists than across lists, analogous to behavior at event boundaries in event memory tasks (such as watching and remembering a film with narrative shifts). Movie clips from Davis et al.²⁵.

To address this gap, we build a computational model that organizes and searches memories hierarchically. We achieve this by extending the CMR model to incorporate a hierarchical structure and specifying how it supports efficient retrieval. To illustrate the

model, consider an office meeting that unexpectedly becomes a birthday party (Figure 1). As the experience unfolds, the CMR model maintains a single, item-level context that slowly updates with each new item, like “pencil” and “screen”. Learning then occurs by associating each item with the current state of the context (Figure 1a, yellow ovals). When an event boundary occurs (e.g., a “cake” appears), it triggers a sharp contextual shift to the first item of the new event, separating the “meeting” memories from the new “party” memories²⁰. At recall, the current item-level context guides retrieval of memories based on their context similarity. Our hierarchical model builds on this foundation and introduces three additional mechanisms. First, instead of one context, our model maintains multiple levels of context (e.g., item-level and event-level) that evolve at different rates (*Multiple Contexts*; Figure 1a), consistent with how the brain represents information at different timescales. Second, while the item-level and event-level contexts drift at different rates within an event, at event transitions, the item-level context resets and synchronizes to the event-level context (*Boundary Synchronization*; Figure 1a, orange ovals). This way, a representation is encoded for each event in the event-level context, and items like “pencil” and “screen” are learned near their appropriate event-level context of “meeting” and items like “cake” and “balloons” are learned near their event-level context of “party” (Figure 1b, left). Third, these hierarchical representations formed at encoding enable efficient memory search at retrieval. While the event-level context is not directly associated with items, the synchronization process creates a link between the event-level context and item-level context at event boundaries (Figure 1b, left, dashed lines). If retrieval based on the current item-level context fails, one can search across the encoded event-level contexts to reactivate item-level contexts through event boundaries (*Hierarchical Retrieval*; Figure 1b, right). For example, after one can no longer retrieve more party experiences, one can use event-level information to retrieve the “meeting” context, which can drive more recalls, like “pencil”.

To test our model’s ability to account for hierarchical memory search, we examine its predictions in the free recall task and the final free recall task. Free recall allows us to examine typical behavioral patterns at the level of item recall, as only one list is studied and recalled at a time^{16,17} (Figure 1c). Final free recall allows us to examine similar patterns at the level of list recall, as a series of lists are learned and then items from all of the lists are recalled in any order²⁴ (Figure 1d). Thus, we examine if our model captures hierarchical behavior in how one recalls and transitions among items within a list (in free recall) and across lists (in final free recall). Further, the transitions between lists create contextual shifts in the final free recall paradigm²⁰, similar to event boundaries^{26,27}, allowing us to examine recall patterns analogous to those in the event memory literature. This literature has used various paradigms to study how participants perceive and remember continuous experiences and segment them into discrete events (such as watching a series of movie clips, Figure 1e²⁵, reading or listening to a narrative²⁸, or encoding a series of items separated by visual and/or task features^{29–32}). The event boundaries induce selective trade-offs: sequential recall is worse across event boundaries than within events^{28–31} while recall of items at event boundaries is higher²⁹. We will examine if our model predicts similar patterns around list boundaries in final free recall, and further refer to the model’s event-level context as a list-level context.

To foreshadow our results, the hierarchical model unifies a wide range of memory phenomena under a single framework. By tracking contextual changes at multiple levels, it

captures recency and temporal contiguity effects at the item-level in free recall and the list-level in final free recall. Our model also explains enhanced recall for start-of-list items (primacy) in both recent and remote lists, whereby the higher-level, list contexts encoded at boundaries are privileged entry points, providing access to the lower-level item contexts. The hierarchical search mechanism also accounts for patterns in recall amount and transition at list boundaries, analogous to boundary effects observed in event memory tasks. Importantly, our model achieves these results while maintaining model sparsity, using the same number of parameters as the single-level context model CMR. We conclude these findings support our model’s account of how humans hierarchically encode and search their memories.

Results

To examine hierarchical memory behavior in our proposed model, we fit model parameters to the free recall task using data from Kahana et al.³³, and then generalized the same parameter set to the final free recall task to capture qualitative behavioral trends from the PEERS dataset³⁴ and from the event memory literature. We used Bayesian optimization³⁵ to minimize the normalized root-mean-squared error between the model-generated and human behavioral patterns in the free recall dataset. We then qualitatively fit one additional parameter to the final free recall and event memory behavior. With this parameter fit, we simulated 1000 trials, each consisting of free recall of 10 lists of 10 items, followed by final free recall of all 100 items. We then compared our model to free recall data³³ and final free recall data³⁴. Full details on the model parameters, estimation procedure, and datasets are provided in the Methods section.

Similar Recall Patterns Emerge Across Timescales

We first tested if our model captured patterns of recency and temporal contiguity effects in its recall of items and lists. As participants can recall items in any order, patterns at multiple levels suggest temporal organization at multiple timescales³⁶. At the item-level in free recall, participants exhibited (i) recency effects whereby there is enhanced recall of later serial positions (shown on the serial position curve¹⁶; Figure 2b), and (ii) temporal contiguity effects whereby successive recalls tend to come from nearby serial positions (shown on the conditional response probability (CRP) which is computed by dividing the number of times a transition to each lag is actually made by the number of times it could have been made for each serial position²¹; Figure 2c). These patterns are scale-invariant, extending from short intervals within a list to longer intervals across lists. As seen in final free recall, participants displayed (i) enhanced recall of the last few lists (Figure 2d) and (ii) a tendency to transition to nearby lists (Figure 2e; see the Method for details on computing the CRP by list lag). Figure 2a illustrates how these effects manifest at the item-level and list-level in an example recall task of five lists of ten items each. The recency effect is shown by the enhanced recall of later serial positions (9 and 10) and later list positions (4 and 5). The temporal contiguity effect is observed in the tendency to transition to nearby items (+1 lag; from “forest” to “card”) more often than remote items (-8 lag; from “forest” to “cloud”) within a list, and to transition to nearby lists (-1 lag; from list 5 to list 4) more often than remote lists (-4 lag; from list 5 to list 1).

Our proposed hierarchical model captured these scale-invariant patterns. At the item-level, our model displayed recency on the serial position curve (Figure 2f) and

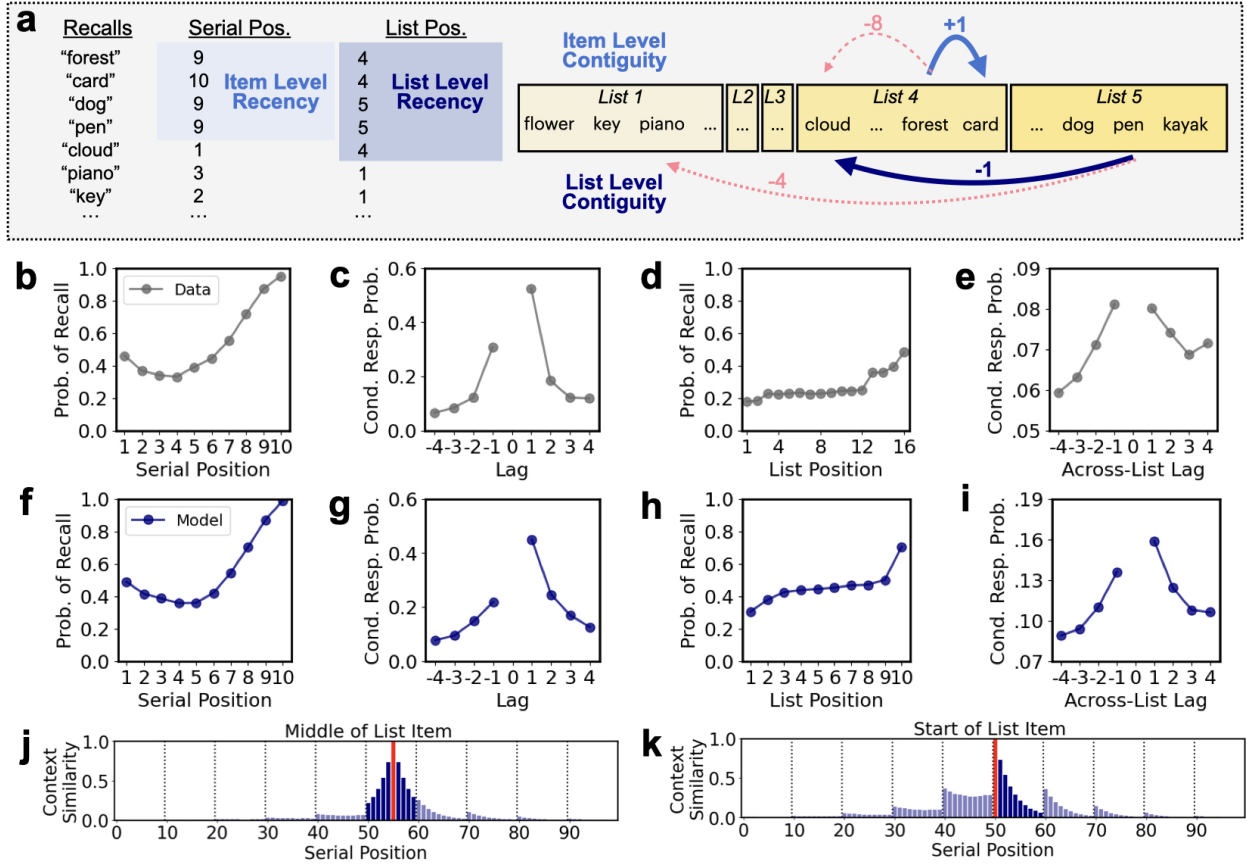


Figure 2

Recall patterns persist across timescales, which our model captured by maintaining similarity between nearby items and lists. (a) Example final free recall output illustrating recency and temporal contiguity at the item-level and list-level. In Kahana et al.'s free recall data³³, item-level effects were observed on the (b) serial position curve and (c) conditional response probability (CRP) by item lag. In the PEERS dataset's final free recall data³⁴, list-level effects were observed on the (d) recall probability by list and (e) CRP by list lag. Our model captured item-level effects on the (f) serial position curve and (g) CRP by item lag. Our model also captured list-level effects on the (h) recall probability by list and (i) CRP by list lag. (j) Due to the drifting item-level context at encoding, an item's context is more similar to the contexts of items studied nearby versus further away within a list. (k) Due to the slower drifting list-level context, a list's context is more similar to the contexts of lists studied nearby versus further away.

temporal contiguity on the CRP by item lag (Figure 2g). These recall patterns are driven by context similarity, an item-level mechanism inherited from the CMR model. Recency arises because the start-of-recall context overlaps the most with the encoding contexts of recently studied items. Temporal contiguity occurs because recalling an item retrieves its encoding context, which in turn cues nearby items with similar encoding contexts. To illustrate the ability of our proposed model in capturing similarity across items within a list, we computed the cosine similarity between the items' encoding contexts. The context

of a mid-list item is more similar to those of nearby items than to remote items within that list due to the drifting item context at encoding (Figure 2j).

Our model displayed the same effects at the list-level, showing recency on the recall probability by list (Figure 2h) and temporal contiguity on the CRP by list lag (Figure 2i). Our model explains these effects through its list-level context. The list-level recency effect arises because at the start of recall, the current list-level context overlaps with the contexts of recently studied lists, supporting the retrieval of the list-level context of these lists. Retrieving a list-level context reactivates its associated item-level context, supporting the retrieval of items in that list. The list-level temporal contiguity occurs because retrieving a list-level context in turn cues nearby lists with similar encoding contexts, supporting the retrieval of the list-level context of these lists, and subsequently items within these lists. To illustrate the ability of our proposed model in capturing context similarity across lists, Figure 2k shows that the item-level context of a start-of-list item is more similar to contexts of nearby lists than remote lists. Because the item-level context resets to the list-level context at list boundaries, the similarity observed across the start-of-list items reflects the underlying list-level context similarity. Taken together, our model captures recall patterns across multiple time scales by tracking two context states that drift at different rates: an evolving item-level context that maintains similarity between items, and a slower list-level context that maintains similarity between lists.

Primacy Effects as Entry Points During Hierarchical Retrieval

An effective memory system must not only track experiences at different timescales, but also search them efficiently by first perceiving them as discrete events and later accessing these events. The free recall and final free recall tasks allow us to examine how human participants encode discrete lists and later retrieve information across them. Critically, we hypothesize that the widely observed primacy effect (enhanced recall of start-of-list items¹⁶) is a result of a hierarchical memory search process attempting to access a particular list before retrieving its associated items. Comparing our model to free recall data from Kahana et al.³³ and final free recall data from the PEERS dataset³⁴, we found primacy effects for both recent lists (in free recall; Figure 3a) and remote lists (in final free recall when averaged across all lists, Figure 3b; and when split by each list, Figure 3d). The data also show a negative recency effect²⁴: end-of-list items are recalled well during free recall (Figure 3a) but poorly during final free recall (Figures 3b, 3d).

Our model captured primacy effects for both recent lists (in free recall) and remote lists (in final free recall) through its hierarchical search processes. Specifically, the item-level and list-level contexts are synchronized at the beginning of encoding each list, so that during recall, if the item-level search fails, the model can search across the encoded list-level contexts as entry points to reactivate a new item-level context to drive further recall. This creates a primacy effect because the reactivated item-level context matches the beginning-of-the-list context, which is most similar to the early list items. In free recall, only one list-level context is available for retrieval, leading to primacy for that recent list (Figure 3e). In final free recall, where there are multiple list-level contexts encoded, the model retrieves the list-level context most similar to the current list-level context, giving rise to primacy effects for remote lists (Figure 3f). The model also captured the negative recency effect, where end-of-list items are recalled well during free recall (Figure 3e) but

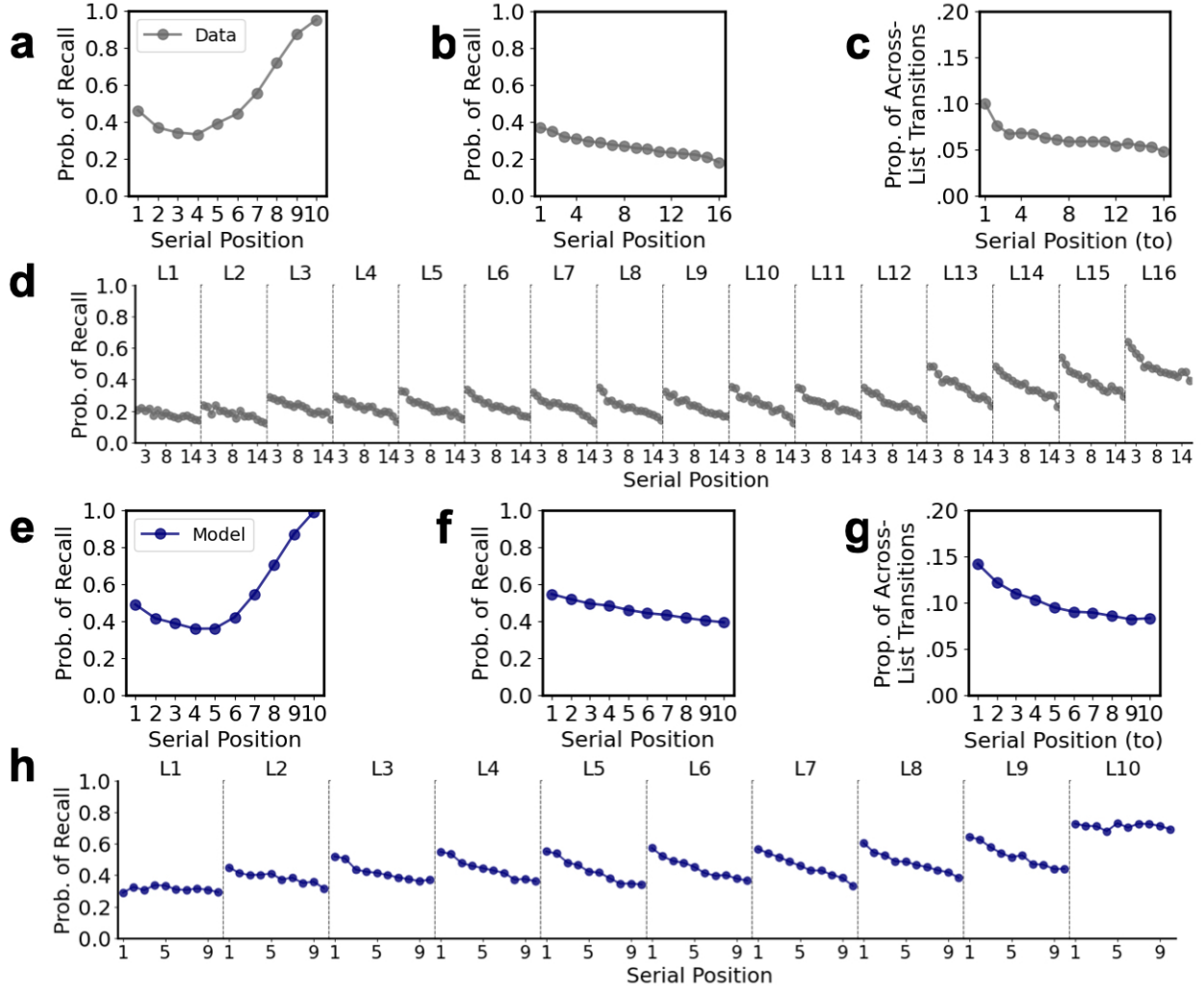


Figure 3

Primacy effects explained by list-level contexts serving as entry points for recall. (a) The serial position curve for free recall data from Kahana et al.³³. (b) The serial position curve for final free recall data from the PEERS dataset³⁴ when averaged across all lists and (d) when split by each list. (c) The probability of retrieving a serial position upon transitioning to a new list during final free recall. Our model predicted similar behavior on serial position curves for (e) free recall and (f) final free recall when averaged across all lists (h) and when split by each list, as well as (g) when transitioning to a new list.

not during final free recall (Figures 3f, 3h). This is because items in the most recent list can be accessed through both the beginning-of-list context during list-level context retrieval and the end-of-list context which matches the start-of-recall context; however, items in the remote lists can only be accessed through the beginning-of-list contexts which are activated during list-level context retrieval.

Beyond demonstrating the alignment between the model behavior and well-established primacy effects, we found further evidence supporting the hierarchical search process. When participants' recalls transitioned to a new list, they tended to enter

the list by recalling an item from the list’s beginning (Figure 3c), an effect our model also predicted (Figure 3g). The evidence of participants entering at lists’ beginnings suggests that they maintain higher-level information about the lists’ onset that can be used at retrieval to access the list. Taken together, these results support our model’s hierarchical search mechanism, which retrieves a start-of-list context as an entry point to resume recall when the current context fails. Our results reveal that these effects are emergent properties of a single, context-driven, hierarchical search mechanism.

List Boundary Effects Mirror Event Boundary Effects

Finally, we examined if our model captured similar behavior around list boundaries in final free recall as observed around event boundaries in event memory tasks³⁷. In those tasks, participants perceive and remember continuous experiences by segmenting them into discrete events^{25,28,30}), and demonstrate enhanced recall for information at event boundaries (Figure 4a), similar to primacy effects in free recall. People also demonstrate better memory within versus across events, with better sequential recall and more transitions within events (Figure 4b). Furthermore, transitions rarely cross event boundaries: after recalling an event’s last item, people tend to stay within that event (Figure 4c), and after recalling an event’s first item, they tend to move forward within the list rather than backward to the previous one (Figure 4d). Since these specific boundary dynamics have not been examined in final free recall, we tested whether our model would predict analogous behaviors when treating lists as events.

Our model aligned with human behavior at event boundaries. First, we examined if boundary information is recalled better than non-boundary information, as observed in naturalistic studies of free recall³⁸ and recognition³⁹ (Figure 4e). Our model predicted significantly greater recall of the first item of each list versus the average recall of all other items (Wilcoxon signed-rank test: $W = 78598.5$, $n = 1000$, $p < .001$, two-tailed; Figure 4i). This enhanced recall for boundary information is evidence for hierarchical retrieval, as event boundaries are retrieved first before they are used to access information within the event. Next, we examined whether the model shows stronger memory associations within events than across them. This has been observed multiple ways (Figure 4f); for instance, participants more accurately recognized a pair of experiences from the same event than different events (Wang & Egner, 2022⁴⁰, “W&E”; DuBrow & Davachi, 2014⁴¹, “D&D”), recalled a subsequent experience better when the cued experience was from the same event than from a different event (Horner et al., 2016⁴², “H+”), and clustered their free recall of items within events more than by chance (Polyn et al., 2009⁴³, “P+”). Our model aligned with this behavior in final free recall, with the proportion of within-list transitions being significantly greater than when each trial’s recall was randomly shuffled (Wilcoxon signed-rank test: $W = 3246.0$, $n = 1000$, $p < .001$, two-tailed; Figure 4j). Finally, we analyzed transitions made at event boundaries. While recall transitions are biased in a forward direction in free recall²¹, prior work has found evidence of participants being less likely to transition across boundaries after remembering an item just before the boundary^{30–32}. Specifically, in free recall tasks, after recalling the last item in an event, participants were more likely to transition backwards to an earlier item within the list than forwards to an item in the next list³² (Figures 4c, 4g). Similarly, after recalling the first item in a list, participants were more likely to transition forwards to other items within the

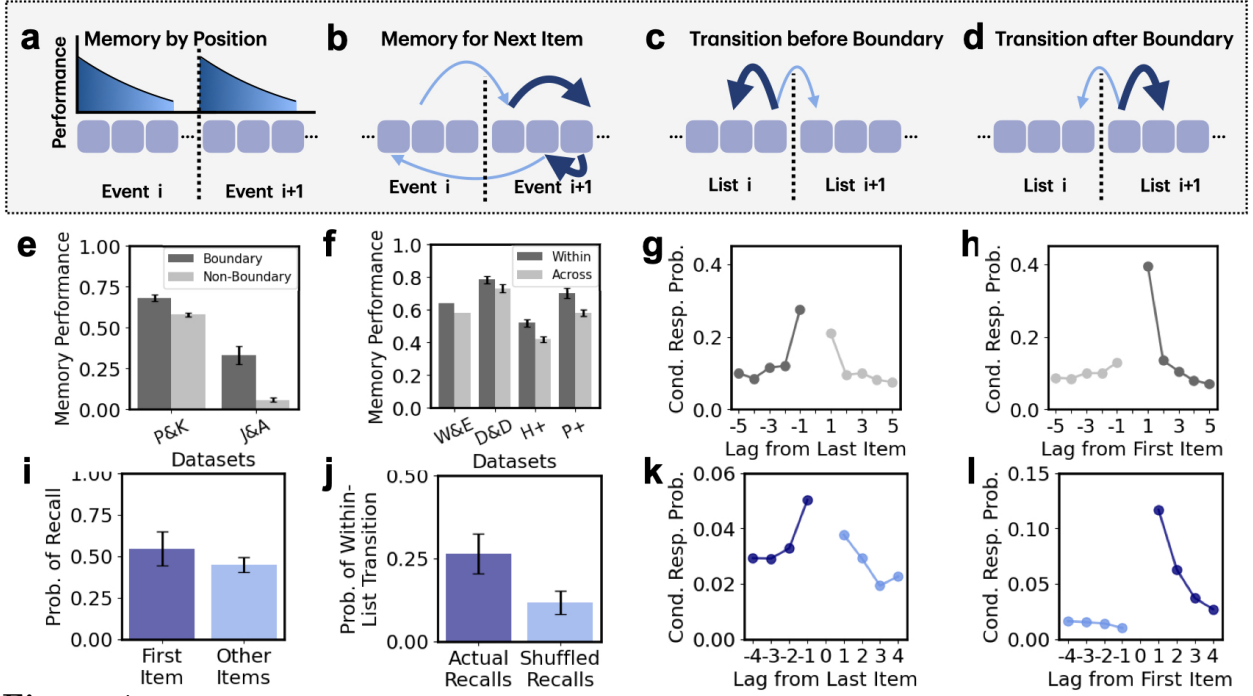


Figure 4

Our model predicted effects of list boundaries analogous to observed effects at event boundaries. The event memory literature finds: (a, e) better memory for boundary versus non-boundary information in narrative recall tasks (reproduced from Figure 2 in Pradhan & Kumar, 2022³⁸, Exp. 1 and from reported results in Jeunehomme & D’Argembeau, 2020³⁹); (b, f) stronger memory for items within the same event than across events, an effect found in recency discrimination (reproduced from Figure 3A in Wang & Egner, 2022⁴⁰, Exp. 1 and Figure 2 from DuBrow & Davachi, 2014⁴¹, Exp. 1), cued recall (reproduced from Table 1 in Horner et al., 2016⁴², Exp. 1), and free recall tasks (reproduced from Table 1 in Polyn et al., 2009⁴³); (c, g) a greater likelihood of backward recall transitions after recalling the last item in an event³²; and (d, h) a greater likelihood of forward recall transitions after recalling the first item in an event³². Correspondingly, our model predicted (i) enhanced memory for the first item in a list compared the average of all other list items, (j) a higher probability of within-list transitions compared to a shuffled baseline, and an increased probability of a subsequent within-list transition after recalling (k) the last item in a list or (l) the first item in a list. Error bars show the standard error of the mean.

list than backwards to an item in the prior list³² (Figures 4d, 4h). Our model captured both of these tendencies to stay within a list at its boundaries (Figures 4k, 4l). These findings support a hierarchical organization in our model, where information from the same event is encoded into a more proximate context space than information from different events. Taken together, these results demonstrate our model’s ability to unify empirical patterns from both list learning paradigms and event memory tasks under a single hierarchical framework.

Discussion

Prior work supports the existence of hierarchical representations in the brain, but a formal theory of how humans search this structure is lacking. To address this gap, we built a hierarchical model of memory search by extending the Context Maintenance and Retrieval (CMR) model^{19,20} and specifying efficient encoding and retrieval processes. We evaluated our model against a free recall dataset²¹ for single-list recall and a final free recall dataset³⁴ for multi-list recall. Our model captured memory search behaviors in both the list learning literature and event memory literature. Specifically, it reproduced scale-invariant effects of recency and temporal contiguity at both the item-level in free recall and the list-level in final free recall. Our model also provided a single, context-based account of primacy effects for recent and remote lists. Furthermore, our model accounted for patterns in recall amount and transition around list boundaries, similar to those observed in the event memory literature. These results support our model’s account of efficient hierarchical memory search, which operates by tracking multiple levels of contexts and allowing higher-level context to reactive lower-level item information. We now turn to the broader implications of our model.

While prior models of memory search have focused on single-list recall^{18–20}, we propose a hierarchical model that accounts for behavior in both single-list and multi-list paradigms. Our model achieves this by tracking two contextual states that drift at different rates: an item-level context that maintains similarity between items, and a slower, list-level context that maintains similarity between lists. This hierarchical representation allows the model to capture scale-invariant effects of recency and temporal contiguity at both the item and list levels. Our approach aligns with neuroscience findings of hierarchical processing in the cortex, with low-level brain regions tracking fast changes and high-level regions tracking slow changes^{9–12}. A model by Howard et al.⁴⁴ also proposed the tracking of multiple contextual states – through a set of leaky integrators with a spectrum of time constants⁴⁴. However, learning in their model requires associating every item with every level of context at each timestep. In comparison, our hierarchical model has a more parsimonious memory representation, associating items with only the item-level context, and associating list-level contexts with item-level contexts only at list boundaries. Further, an effective memory system organizes a continuous stream of information into discrete, searchable events. While Howard et al.’s⁴⁴ model assumes a continuous stream of stimuli as input, our model additionally accounts for effects associated with memory access at list boundaries. It does so by synchronizing the item-level context with the list-level context at list boundaries, and later using these encoded list-level contexts to reactivate memories. Thus, we provide an efficient account of episodic memory search by integrating discrete list structures with multiscale contextual representations.

A robust finding in the memory literature is enhanced recall at list boundaries for start-of-list items (primacy effects)¹⁶ and enhanced recall at event boundaries^{38,39}. Prominent models like the Context Maintenance and Retrieval (CMR) model capture primacy for a recently studied list by resorting to separate mechanisms – such as increased attention to early list items – that are distinct from the core process of context-driven memory search. Other theories have similarly invoked specific mechanisms. For example, early theories focused on increased rehearsal for the early items in the list, either from

increased encoding in a short-term memory buffer⁴⁵ or from the repetition and distribution of rehearsals⁴⁶. Alternatively, the perceptual or temporal distinctiveness for early list items has also been suggested as a mechanism for primacy^{17,47}. Other accounts have posited that a shift in context at a list’s onset enhances their encoding via novelty⁴⁸ or prediction error²⁶. Moreover, primacy for remote lists has often been accounted under different mechanisms than those that explain primacy for a recently studied list, whereby one retrieves a list tag, representing either an entire list^{49–51} or the beginning-of-list context⁵². Without assuming separate mechanisms to account for primacy effects or to differentiate between primacy for recent and remote lists, here, we explain these effects as emergent properties of a single, context-driven, hierarchical search mechanism. If the item-level memory search fails, the model can use the encoded list-level contexts as new entry points. This reactivates the item-level context from a list’s beginning, enhancing recall of early list items. In free recall, only one list context is available, leading to primacy for that recent list. In final free recall, the model retrieves the list context most similar to its current state, allowing it to produce primacy effects for remote lists.

Beyond providing a unified mechanism for primacy effects, our model further explains why maintaining access to the beginning of a list or event is functionally adaptive. While the mechanism of “jumping back” to the start of a list exists in other models^{52,53}, our model deploys this action purposefully when the current search is exhausted, allowing it to navigate across different levels of information. Past theoretical work has demonstrated why it is optimal to enter a list through its beginning rather than elsewhere in the list: initiating recall from the beginning of a list, rather than other list positions, contributes to improved memory performance of that list⁵⁴. Furthermore, the functional role of list boundaries aligns with prior work in the event memory literature. For instance, the presence of an event boundary can increase the overall recall amount in lists of words and narrative texts⁵⁵. People have also been shown to use event boundaries to flexibly navigate their experiences. In one study, the time it took participants to recall a moment in a video clip, after being cued with a prior moment, was better explained by the number of events they had to skip between the moments, rather than the actual interval duration⁵⁶. In another study, participants’ recall appeared to proceed hierarchically, recalling first scenes, then sub-scenes of the movie⁵⁷. These results are consistent with a mechanism where people search for an access point, followed by a finer scan of information within an event before transitioning to a new event⁵⁶. Our model complements this body of work by demonstrating how enhanced recall at boundaries can directly emerge from a hierarchical search mechanism.

The notion that continuous experiences are divided into discrete events is a fundamental idea in cognitive science. There are many ways to define events⁵⁸, such as through subjective ratings⁶, neural representation¹², perception⁵⁹, or the presence of contextual shifts^{26,27}. In the present work, we evaluate our model’s ability to capture key effects from the event memory literature by treating separate word lists from free recall experiments as individual events. This is justified because the transitions between lists create contextual shifts²⁰, a known marker of event boundaries^{26,27}. While many other important features characterize an event, using a well-controlled list learning paradigm – and increasing its empirical complexity from single-level to hierarchical materials – provides a tractable framework for extending the existing theory of memory search to

incrementally more naturalistic scenarios⁶⁰. Our model successfully accounts for recall and transition patterns at list boundaries that are analogous to those at event boundaries, thereby providing a theoretical unification of findings from list learning and event memory research. However, our model does not address how these boundaries are identified from rich, continuous stimuli important in real-world experiences. A central feature of naturalistic event memory is the capacity to segment continuous experience into discrete events^{2,7}. Future theoretical work should also account for this segmentation process. The development of computational theories for naturalistic event memory is an area of active research. A promising direction is to combine a hierarchical search framework like ours with an event segmentation mechanism for naturalistic stimuli⁶¹, methods to identify event boundaries in human fMRI data¹², or large language models that can predict event boundaries by measuring Bayesian surprise⁶². It remains an intriguing future direction for a computational theory to both infer the event structure of rich stimuli and model the detailed memory search of events.

In summary, we present a hierarchical model of memory search by organizing memories at multiple levels and using higher-level contextual information as cues to activate lower-level item information. Our model is capable of accounting for memory search behavior in free recall, final free recall, and event memory paradigms under a single framework. The alignment between our model and the empirical behavior provides support for our model’s efficient account of how humans hierarchically encode and search their memories.

Method

In this section, we first review the Context Maintenance and Retrieval (CMR) model which was developed to account for behavioral patterns in the free recall task^{19,20}. We then introduce our model of hierarchical memory search which extends the CMR model to account for patterns in both the final free recall task as well as the free recall task. We discuss our model simulation details and parameter estimation method, in which we assume parameters fit to free recall generalize to final free recall. Finally, we overview the datasets to which we fit model parameters and compare model predictions.

The Context Maintenance and Retrieval Model (CMR)

As participants study lists of items in the free recall task (a total of N words), CMR proposes that their item-level context drifts towards the memory representations of recently encountered experiences. When an item is presented in the study list, the state of the item-level context at timestep i ($c_i \in \mathbb{R}^N$) is given by:

$$c_i = \rho c_{i-1} + \beta_{enc} c_{enc}^{IN} \quad (1)$$

where $c_{enc}^{IN} \in \mathbb{R}^N$ is the pre-experimental context induced by the presented item, $\beta_{enc} \in [0, 1]$ is a parameter determining the rate at which context drifts toward the presented item, and ρ is a scalar ensuring $\|c_i\| = 1$. The pre-experimental context is expressed as $c_{enc}^{IN} = M_{pre}^{FC} f_i$, where $M_{pre}^{FC} \in \mathbb{R}^{N \times N}$ represents item-to-context associations that existed prior to the experiment and f_i is a binary vector that is all zeros except at the presented item’s position. For simplification, M_{pre}^{FC} is initialized as an identity matrix, assuming that an item is only associated with its own context.

In addition to updating the item-level context, associations form between the presented item f_i and the evolving item-level context state c_{i-1} to capture new learning. These associations are held in two matrices ($M_{exp}^{FC} \in \mathbb{R}^{N \times N}$ and $M_{exp}^{CF} \in \mathbb{R}^{N \times N}$) that are initialized to zero:

$$\Delta M_{exp}^{FC} = c_{i-1} f_i^T \quad (2a)$$

$$\Delta M_{exp}^{CF} = f_i c_{i-1}^T \quad (2b)$$

The overall effect of the item-level context drifting towards the pre-experimental context of presented items in Equation 1 and associative learning in Equation 2 is that each item is embedded at a location in the item-level context space near other recently presented items. As items studied nearby in the list are tied to similar contextual states during encoding, subsequent recalls are likely to be nearby items in the study list, leading to *temporal contiguity effects*.

At recall, the item-level context drifts towards the context induced by a just-recalled item, similarly to how context drifts during encoding in Equation 1. The state of the item-level context at recall timestep j is given by:

$$c_j = \rho c_{j-1} + \beta_{rec} c_{rec}^{IN} \quad (3)$$

where $\beta_{rec} \in [0, 1]$ determines the drift rate at recall and ρ is a scalar ensuring $\|c_j\| = 1$. Different than encoding, $c_{rec}^{IN} \in \mathbb{R}^N$ is now comprised of an item's pre-experimental context ($M_{pre}^{FC} f_j$) as well as its experimental context formed during study ($M_{exp}^{FC} f_j$):

$$c_{rec}^{IN} = (1 - \gamma_{fc}) M_{pre}^{FC} f_j + \gamma_{fc} M_{exp}^{FC} f_j \quad (4)$$

where the relative weight is determined by a parameter, $\gamma_{fc} \in [0, 1]$.

Once context drifts towards this retrieved context, which items are likely to be recalled? The activation strength a at timestep j for recalling different items depends on how much the current context c_j matches with items' experimental contexts stored in M_{exp}^{CF} :

$$a = \phi_l M_{exp}^{CF} c_j \quad (5)$$

Here, M_{exp}^{CF} is scaled by ϕ_l to simulate increased attention to start-of-list items, allowing the model to capture *primacy effects* (i.e., enhanced recall for early list items). Specifically, $\phi_l = \phi_s e^{-\phi_d(l-1)} + 1$ is an exponential decay function, creating a primacy gradient where items at early serial position l receive higher attention at recall. Given the activation strength, we used the softmax function as the retrieval rule⁵⁴: the probability of recalling the item from encoding timestep i , at recall timestep j is given by:

$$P(f_j = f_i) = \frac{e^{ka_i}}{\sum_{i'=1}^N e^{ka_{i'}}} \quad (6)$$

where the parameter k is the amount of noise at retrieval. As one begins recall by using their end-of-study context to guide search, end-of-list items are likely to be recalled, giving rise to *recency effects*. Once an item is retrieved, the item-level context state drifts towards

the just-recalled item; this updated context state supports the retrieval of a new item. This retrieval process continues until determined by the stopping rule:

$$p_j^{stop} = e^{-\epsilon_d a^{nr}/a^r} \quad (7)$$

where a^r indicates the summed activation of already-recalled items, a^{nr} indicates the summed activation for not-yet-recalled items, and ϵ_d is a scaling parameter^{53,54}.

Finally, in typical free recall experiments, a participant often repeats the process of encoding and then recalling the just-studied list for multiple trials. After the retrieval of one list ends, the model posits a sudden shift in the item-level context to account for the change in list, separating items in the new list from items in the previous list. A new item is presented following Equation 1, and the parameter β_{post} determines the rate of this context shift. The item is not associated with item-level context, and thus, is not represented in the associative matrices nor enters into retrieval. In total, the CMR model has eight parameters for simulating free recall (see Table 1).

Incorporating Hierarchical Memory into CMR

We extend the CMR framework to capture hierarchical behavior in the final free recall task in which a series of M lists are learned and then items from all of the lists are recalled in any order²⁴. To simulate the final free recall task, we build upon the CMR framework by introducing three additional mechanisms: (i) *Multiple Contexts*, (ii) *Boundary Synchronization*, and (iii) *Hierarchical Retrieval*. We explain each of these below.

First, the model tracks multiple levels of context. In addition to an item-level context that drifts towards the memory representations of recently encountered items posited by CMR, the model also maintains a list-level context that evolves at a slower rate. As items are presented, both context states update towards the pre-experimental context of each item, c_{enc}^{IN} . At timestep i , the item-level context is given by Equation 1 and the list context ($c_i^{list} \in \mathbb{R}^N$) is given by:

$$c_i^{list} = \rho c_{i-1}^{list} + \beta_{list} c_{enc}^{IN} \quad (8)$$

where $c_{enc}^{IN} \in \mathbb{R}^N$ is the same retrieved, pre-experimental context induced by a presented item in Equation 1, and ρ is a scalar ensuring $\|c_i^{list}\| = 1$. A different drift rate parameter, $\beta_{list} \in [0, 1]$ where $\beta_{list} < \beta_{enc}$, allows the list-level context to evolve slower than the item-level context.

Second, the item-level context synchronizes with the list-level context at list boundaries. As one moves from one list to the next in final free recall, the item-level context resets to the state of the list-level context:

$$c_i^{item} = c_i^{list} \quad (9)$$

Then, the state of the list-level context is encoded into a matrix ($M^{lists} \in \mathbb{R}^{M \times N}$) that is initialized to zero:

$$\Delta M^{lists} = g_m c_i^{list\top} \quad (10)$$

where g_m is a binary vector with all zeros except at the presented list's position,

represented by the index m .

The process of (i) maintaining two context states drifting at different rates (Equations 1 & 8) enables the model to capture similarity at the level of both items and lists. The process of (ii) synchronizing the item-level context to the list-level context (Equation 10) creates entry points for later recall, where list-level information can be used to activate item-level information. Together, *list-level recency effects* arise because at the start of recall, the current list-level context is most similar to the contexts of recently studied lists, supporting the retrieval of the list-level context of these lists, and subsequently items within these lists. *List-level temporal contiguity effects* occur because retrieving a list-level context in turn cues nearby lists with similar encoding contexts.

Third, these representations formed at encoding enable the model to have an efficient, hierarchical retrieval. During recall, at timestep j , both the item-level context and list-level context drift towards the context induced by the just-recalled item, c_{rec}^{IN} (defined in Equation 4). The item-level context state is given by Equation 3 and the list-level context state, c_j^{list} , is given by:

$$c_j^{list} = \rho c_{j-1}^{list} + \beta_{list} c_{rec}^{IN} \quad (11)$$

where β_{list} maintains the same drift rate as during encoding (Equation 8) and ρ is a scalar ensuring $\|c_j^{list}\| = 1$.

Once the item-level and list-level contexts are updated, which items are likely to be recalled? Our model starts the retrieval by following a similar process to CMR, whereby the item-level context drives recall following Equation 5, with the critical difference that M_{exp}^{CF} is not scaled by a primacy gradient $\phi_l = \phi_s e^{-\phi_d(l-1)} + 1$ that increases the activation of early list items at serial position l (thus removing two parameters ϕ_s and ϕ_d from the base CMR model). Items are retrieved following Equation 6 until determined by the same stopping rule (Equation 7). Then, when the item-level context is no longer an effective cue for retrieval, the list-level context can be used to reactivate the item-level context at list boundaries to continue recall. To determine which list-level context to reinstate, the activation strength, a^{list} at timestep j , for recalling different list-level contexts depends on the similarity between the current list-level context, c_j^{list} , and the list-level contexts held in M^{lists} :

$$a^{list} = M^{lists} c_j^{list} \quad (12)$$

Given the activation strength, we used the softmax function to determine the probability of retrieving the list-level context of list position m at recall timestep j :

$$P(g_j = g_m) = \frac{e^{ka_m^{list}}}{\sum_{m'=1}^M e^{ka_{m'}^{list}}} \quad (13)$$

where the parameter k determines the amount of noise (inherited from item-level retrieval in Equation 6). Once a list-level context is retrieved, the current list-level context state,

c_{j-1}^{list} , updates towards the retrieved list-level context, c_{rec}^{list} :

$$c_j^{list} = \rho c_{j-1}^{list} + \beta_{rein} c_{rec}^{list} \quad (14)$$

where, following the list-level context drifting process in Equation 8, the list-level context drifts towards the retrieved list-level context at a rate determined by the parameter $\beta_{rein} \in [0, 1]$ and ρ is a scalar ensuring $\|c_j^{list}\| = 1$. Then, the item-level context is reactivated to the list-level context:

$$c_j^{item} = c_j^{list} \quad (15)$$

This newly updated, item-level context state is used to drive further recall (Equations 5–6) until there is insufficient support at the item-level retrieval (Equation 7), after which the model again resorts to the list-level context. This process iterates until there are no more lists to be retrieved. *Primacy effects* arise in the model because the reactivated item-level context (which is identical to the list-level context at list boundaries) matches the beginning-of-the-list context that is most similar to the early list items. This places primacy, recency, and temporal contiguity effects under the same context-driven, hierarchical search mechanism.

Beyond the final free recall task, the model can also capture the free recall task, as free recall is considered a special case of final free recall, where there is only one target list to be retrieved. Therefore, the free recall task should share the same processes and model parameters as the final free recall task, with the only exception that the retrieval of the list-level context, once the item-level context fails, is simply set to be the beginning-of-list context instead of through a list-level retrieval process (i.e., removing Equations 12–13 and the parameter β_{post}). Overall, although our model introduces additional processes at the hierarchical level, it maintains model sparsity, without increasing the total number of parameters compared with that of CMR. Six parameters are inherited from the CMR model, and two new parameters characterize the list-level processes (see Table 1).

Model Fitting and Simulations

As the same memory search processes govern free recall and final free recall in our model, we fit the model parameters to free recall and evaluate the model’s ability to generalize to final free recall behavior. We used free recall data from Kahana et al.’s³³ Experiment 1 ($N = 31$ young adults), and fit the model across three behavioral patterns: (1) the probability of recall by serial position, (2) the probability of first recall by serial position, and (3) the conditional response probability (CRP) at serial position lags -4/+4. We used 200 iterations of Bayesian optimization³⁵, after 400 random initializations, to obtain a set of parameters that minimized the root mean squared error between the empirical and simulated values across the three patterns. Table 1 reports the model parameter fit for free recall. With this parameter fit, we simulated 1000 trials, each consisting of free recall of 10 lists of 10 items, followed by final free recall of all 100 items.

To fully simulate final free recall behavior, our model requires one additional parameter, β_{post} , that describes the amount of context change between each list (see Table 1). We searched for a value from 0 to 1 across nine final free recall behaviors. Each pattern is expanded upon in the Results section, but we briefly name them here: recall by list

Table 1. Model Parameters

	Parameter Name	CMR Base	Free Recall	Final Free Recall	Description
Our Model	β_{list}	×	0.400	0.400	Rate of list-level context drift
	β_{rein}	×	0.300	0.300	Rate of list-level context reactivation at retrieval
CMR Model	β_{enc}	✓	0.679	0.679	Rate of item-level context drift at encoding
	β_{rec}	✓	0.326	0.326	Rate of item-level context drift at retrieval
	γ_{fc}	✓	0.315	0.315	Relative weight of retrieving experimental versus pre-experimental context of an item at retrieval
	k	✓	6.50	6.50	Retrieval noise
	ϵ_d	✓	1.04	1.04	Baseline activation for stopping probability
	β_{post}	✓	-	0.850	Rate of list-level context shift recalling each list
	ϕ_s	✓	-	-	Primacy parameter (not assumed in our model)
	ϕ_d	✓	-	-	Primacy parameter (not assumed in our model)

position (Figure 2d), the conditional response probability by list (Figure 2e), the serial position curve for each list (Figure 3d) and averaged across lists (Figure 3b), the serial position curve only including transitions to new lists (Figure 3c), the memory performance for boundary versus non-boundary items (Figure 4e), the memory performance for information within events versus across events (Figure 4f), and the conditional response probability by item lag when transitioning from the last item in a list (Figure 4g) and first item in a list (Figure 4h). We found $\beta_{post} = 0.85$ qualitatively captured these behaviors.

Among these behaviors, we computed the conditional response probability (CRP), which measures the tendency to successively recall information studied at nearby positions. At the item-level, the CRP was computed by dividing the number of times a transition to each lag is actually made by the number of times it could have been made for each *serial* position²¹ (as seen in Figures 2c and 2g). To consider item recalls only made from the end of the list or from the start of the list, we followed the same calculations while only including transition pairs originating from the last serial position (Figure 4k) or first serial position (Figure 4l), respectively. At the list-level (Figure 2e and 2i), the CRP was computed by dividing the number of times a transition to each lag is actually made by the number of times it could have been made for each *list* position. Additionally, following prior work, we only included transition pairs consisting of correct recalls from one list to another list²², and we further controlled for the number of items remaining in each list for each transition pair³⁶.

Free Recall and Final Free Recall Datasets

Kahana et al. (2002) Dataset

We fit our model to a free recall dataset from Experiment 1 by Kahana et al. (2002)³³. Data came from 31 undergraduate students (aged 17–21) who studied 33 lists of words in an immediate free recall task (with the first three lists as practice rounds and excluded from the analysis). Lists were composed of 10 common, two-syllable nouns, and words did not repeat across lists within a participant. At the start of each trial, a fixation cross appeared for 1,400 milliseconds (ms) followed by a blank inter-stimulus interval for 100 ms. During the list presentation, participants were told to say each word aloud.

Immediately following the list, participants had 30 seconds to recall as many items from the list as possible. It was made clear that they did not need to attempt to recall items in the order of presentation. Each session lasted around 90 minutes.

PEERS Dataset

We compare our model behavior to final free recall data from the Penn Electrophysiology of Encoding and Retrieval Study (PEERS), a large database characterizing the electrophysiological correlates of memory encoding and retrieval³⁴. The present analysis is based on behavioral data from the 151 participants who completed Experiments 1 through 3 of the PEERS dataset (aged 18–30) which involved multi-session immediate free recall tasks. Experiment 1 included seven free recall sessions (the first of which was a practice round and excluded from the analysis) and Experiment 3 included six free recall session. For each session, participants studied and immediately recalled a series of 16 lists comprised of 16 common, uncategorized words each. Some lists had words presented concurrently with an encoding task (size judgment, animacy judgment, or both) during which participants responded with a key press. Encoding tasks were counterbalanced across all sessions and participants. Immediately after learning each list, participants were given 75 seconds to freely recall items from the just-studied list. Of the 12 experimental sessions, six sessions included a final free recall task at the end. Participants were given five minutes to recall as many items as possible from the 16 lists. The sessions in Experiment 3 differed from Experiment 1 in that participants were asked to verbalize their recall, but we aggregated the data as the procedure had no reliable effect on correct recall rates²⁰.

References

- [1] Christopher A Kurby and Jeffrey M Zacks. “Segmentation in the perception and memory of events.” In: *Trends in cognitive sciences* 12.2 (2008), pp. 72–79.
- [2] David Clewett, Sarah DuBrow, and Lila Davachi. “Transcending time in the brain: How event memories are constructed from experience.” In: *Hippocampus* 29.3 (2019), pp. 162–183.
- [3] E Tulving. “Episodic and semantic memory.” In: *Organization of memory*. Ed. by E Tulving and W. Donaldson. New York: Academic Press, 1972, pp. 381–402.
- [4] Thomas Suddendorf, Donna Rose Addis, and Michael C Corballis. “Mental time travel and the shaping of the human mind.” In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 364.1521 (2009), pp. 1317–1324.
- [5] E. Tulving. *Elements of episodic memory*. Oxford, UK: Clarendon Press, 1983.
- [6] Darren Newton. “Attribution and the unit of perception of ongoing behavior.” In: *Journal of personality and social psychology* 28.1 (1973), p. 28.
- [7] Jeffrey M Zacks and Barbara Tversky. “Event structure in perception and conception.” In: *Psychological bulletin* 127.1 (2001), pp. 3–21.
- [8] Jeffrey M Zacks, Barbara Tversky, and Gowri Iyer. “Perceiving, remembering, and communicating structure in events.” In: *Journal of experimental psychology: General* 130.1 (2001), pp. 29–58.
- [9] Uri Hasson et al. “A hierarchy of temporal receptive windows in human cortex.” In: *Journal of neuroscience* 28.10 (2008), pp. 2539–2550.
- [10] Christopher J Honey et al. “Slow cortical dynamics and the accumulation of information over long timescales.” In: *Neuron* 76.2 (2012), pp. 423–434.
- [11] Yulia Lerner et al. “Topographic mapping of a hierarchy of temporal receptive windows using a narrated story.” In: *Journal of neuroscience* 31.8 (2011), pp. 2906–2915.
- [12] Christopher Baldassano et al. “Discovering event structure in continuous narrative perception and memory.” In: *Neuron* 95.3 (2017), pp. 709–721.
- [13] Linda Geerligs et al. “A partially nested cortical hierarchy of neural states underlies event segmentation in the human brain.” In: *elife* 11 (2022), e77430.
- [14] Eva Pastalkova et al. “Internally generated cell assembly sequences in the rat hippocampus.” In: *Science* 321.5894 (2008), pp. 1322–1327.
- [15] Daniel M Salz et al. “Time cells in hippocampal area CA3.” In: *Journal of Neuroscience* 36.28 (2016), pp. 7476–7484.
- [16] Bennet B Murdock. “The serial position effect of free recall.” In: *Journal of Experimental Psychology* 64.5 (1962), pp. 482–488.
- [17] Bennet B Murdock. “The distinctiveness of stimuli.” In: *Psychological Review* 67.1 (1960), pp. 16–31.

- [18] Marc W Howard and Michael J Kahana. “A distributed representation of temporal context.” In: *Journal of Mathematical Psychology* 46.3 (2002), pp. 269–299.
- [19] Sean M Polyn, Kenneth A Norman, and Michael J Kahana. “A context maintenance and retrieval model of organizational processes in free recall.” In: *Psychological Review* 116.1 (2009), pp. 129–156.
- [20] Lynn J Lohnas, Sean M Polyn, and Michael J Kahana. “Expanding the scope of memory search: Modeling intralist and interlist effects in free recall.” In: *Psychological Review* 122.2 (2015), pp. 337–363.
- [21] Michael J Kahana. “Associative retrieval processes in free recall.” In: *Memory & Cognition* 24.1 (1996), pp. 103–109.
- [22] Marc W Howard, Tess E Youker, and Vijay S Venkatadass. “The persistence of memory: Contiguity effects across hundreds of seconds.” In: *Psychonomic Bulletin & Review* 15.1 (2008), pp. 58–63.
- [23] Nash Unsworth. “Exploring the retrieval dynamics of delayed and final free recall: Further evidence for temporal-contextual search.” In: *Journal of Memory and Language* 59.2 (2008), pp. 223–236.
- [24] Fergus IM Craik. “The fate of primary memory items in free recall.” In: *Journal of Verbal Learning and Verbal Behavior* 9.2 (1970), pp. 143–148.
- [25] Emily E Davis et al. “Looking the same, but remembering differently: Preserved eye-movement synchrony with age during movie watching.” In: *Psychology and Aging* 36.5 (2021), pp. 604–615.
- [26] Nina Rouhani et al. “Reward prediction errors create event boundaries in memory.” In: *Cognition* 203 (2020), p. 104269.
- [27] Tanya Wen and Tobias Egner. “Retrieval context determines whether event boundaries impair or enhance temporal order memory.” In: *Cognition* 225 (2022), p. 105145.
- [28] Youssef Ezzyat and Lila Davachi. “What constitutes an episode in episodic memory?” In: *Psychological science* 22.2 (2011), pp. 243–252.
- [29] Andrew C Heusser et al. “Perceptual boundaries cause mnemonic trade-offs between local boundary processing and across-trial associative binding.” In: *Journal of Experimental Psychology: Learning, Memory, and Cognition* 44.7 (2018), pp. 1075–1090.
- [30] Sarah DuBrow and Lila Davachi. “The influence of context boundaries on memory for the sequential order of events.” In: *Journal of Experimental Psychology: General* 142.4 (2013), pp. 1277–1286.
- [31] Sarah DuBrow and Lila Davachi. “Temporal binding within and across events.” In: *Neurobiology of learning and memory* 134 (2016), pp. 107–114.
- [32] Lynn J Lohnas, M Karl Healey, and Lila Davachi. “Neural temporal context reinstatement of event structure during memory recall.” In: *Journal of Experimental Psychology: General* 152.7 (2023), pp. 1840–1872.

- [33] Michael J Kahana et al. “Age dissociates recency and lag recency effects in free recall.” In: *Journal of Experimental Psychology: Learning, Memory, and Cognition* 28.3 (2002), pp. 530–540.
- [34] Michael J Kahana et al. “The Penn Electrophysiology of Encoding and Retrieval Study.” In: *Journal of Experimental Psychology: Learning, Memory, and Cognition* 50.9 (2024), pp. 1421–1443.
- [35] Jonas Mockus. “The application of Bayesian methods for seeking the extremum.” In: *Towards global optimization 2* (1998), pp. 117–129.
- [36] M Karl Healey, Nicole M Long, and Michael J Kahana. “Contiguity in episodic memory.” In: *Psychonomic bulletin & review* 26.3 (2019), pp. 699–720.
- [37] Sarah DuBrow, MJ Kahana, and AD Wagner. “Event and boundaries.” In: *Oxford handbook of human memory 1* (2024).
- [38] Rujuta Pradhan and Devpriya Kumar. “Event segmentation and event boundary advantage: Role of attention and postencoding processing.” In: *Journal of Experimental Psychology: General* 151.7 (2022), pp. 1542–1555.
- [39] Olivier Jeunehomme and Arnaud D’Argembeau. “Event segmentation and the temporal compression of experience in episodic memory.” In: *Psychological research* 84.2 (2020), pp. 481–490.
- [40] Yuxi Candice Wang and Tobias Egner. “Switching task sets creates event boundaries in memory.” In: *Cognition* 221 (2022), p. 104992.
- [41] Sarah DuBrow and Lila Davachi. “Temporal memory is shaped by encoding stability and intervening item reactivation.” In: *Journal of Neuroscience* 34.42 (2014), pp. 13998–14005.
- [42] Aidan J Horner et al. “The role of spatial boundaries in shaping long-term event representations.” In: *Cognition* 154 (2016), pp. 151–164.
- [43] Sean M Polyn, Kenneth A Norman, and Michael J Kahana. “Task context and organization in free recall.” In: *Neuropsychologia* 47.11 (2009), pp. 2158–2163.
- [44] Marc W Howard et al. “A distributed representation of internal time.” In: *Psychological Review* 122.1 (2015), pp. 24–53.
- [45] Richard C Atkinson and Richard M Shiffrin. “Human memory: A proposed system and its control processes.” In: *Psychology of learning and motivation*. Vol. 2. Elsevier, 1968, pp. 89–195.
- [46] Lydia Tan and Geoff Ward. “A recency-based account of the primacy effect in free recall.” In: *Journal of Experimental Psychology: Learning, Memory, and Cognition* 26.6 (2000), pp. 1589–1625.
- [47] Ian Neath. “Distinctiveness and serial position effects in recognition.” In: *Memory & cognition* 21.5 (1993), pp. 689–698.
- [48] Eddy J Davelaar. “A novelty-induced change in episodic (NICE) context account of primacy effects in free recall.” In: *Psychology* 4.9 (2013), pp. 695–703.

- [49] Simon Farrell. “Temporal clustering and sequencing in short-term memory and episodic memory.” In: *Psychological review* 119.2 (2012), pp. 223–271.
- [50] Yoonhee Jang and David E Huber. “Context retrieval and context change in free recall: recalling from long-term memory drives list isolation.” In: *Journal of Experimental Psychology: Learning, Memory, and Cognition* 34.1 (2008), pp. 112–127.
- [51] Melissa Lehman and Kenneth J Malmberg. “A global theory of remembering and forgetting from multiple lists.” In: *Journal of Experimental Psychology: Learning, Memory, and Cognition* 35.4 (2009), pp. 970–988.
- [52] M Karl Healey and Christopher N Wahlheim. “PEPPR: A post-encoding pre-production reinstatement model of dual-list free recall.” In: *Memory & Cognition* 52.1 (2024), pp. 163–181.
- [53] James E Kragel, Neal W Morton, and Sean M Polyn. “Neural activity in the medial temporal lobe reveals the fidelity of mental time travel.” In: *Journal of Neuroscience* 35.7 (2015), pp. 2914–2926.
- [54] Qiong Zhang, Thomas L Griffiths, and Kenneth A Norman. “Optimal policies for free recall.” In: *Psychological Review* 130.4 (2023), pp. 1104–1125.
- [55] Kyle A Pettijohn et al. “Event boundaries and memory improvement.” In: *Cognition* 148 (2016), pp. 136–144.
- [56] Sebastian Michelmann, Uri Hasson, and Kenneth A Norman. “Evidence that event boundaries are access points for memory retrieval.” In: *Psychological Science* 34.3 (2023), pp. 326–344.
- [57] James Antony et al. “Causal and chronological relationships predict memory organization for nonlinear narratives.” In: *Journal of cognitive neuroscience* 36.11 (2024), pp. 2368–2385.
- [58] Tristan S Yates, Brynn E Sherman, and Sami R Yousif. “More than a moment: What does it mean to call something an ‘event’?” In: *Psychonomic Bulletin & Review* 30.6 (2023), pp. 2067–2082.
- [59] Brynn E Sherman et al. “Mnemonic content and hippocampal patterns shape judgments of time.” In: *Psychological Science* 34.2 (2023), pp. 221–237.
- [60] Qiong Zhang. “Formal Modeling as Theoretical Glue between Laboratory and Naturalistic Studies of Memory.” In: *Preprint available at: https://osf.io/preprints/psyarxiv/eq8nv_v1* (2025).
- [61] Nicholas T Franklin et al. “Structured Event Memory: A neuro-symbolic model of event cognition.” In: *Psychological review* 127.3 (2020), pp. 327–361.
- [62] Manoj Kumar et al. “Bayesian surprise predicts human event segmentation in story listening.” In: *Cognitive science* 47.10 (2023), e13343.